**SMEB 2207 COMPUTER PROGRAMMING**
**SESSION 2019/2020 SEM 2**
**MINI-PROJECT**

# ELECTRICAL CONDUCTIVITY AND ENERGY GAP FOR SEMICONDUCTORS AND INSULATORS

## THEORY

Electrical conductivity is typically determined by obtaining the current-voltage measurement of a sample of know dimension. The conductivity σ is calculated using the equation:

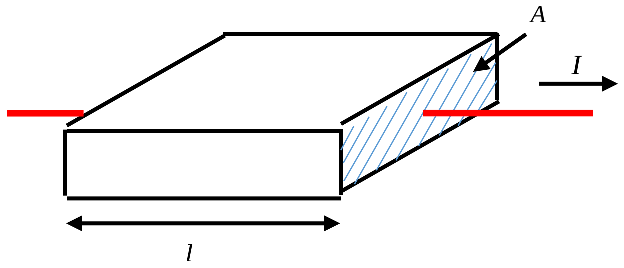
$$\sigma = \frac{I}{V}\frac{l}{A} \qquad (\Omega^{-1}\,\text{cm}^{-1} \quad \text{or} \quad \text{S cm}^{-1})$$

Where $I$ is the current

$V$ is the voltage

$l$ is the length of the material

$A$ is the cross section are of the sample (perpendicular to the direction of the current)




In a current - voltage measurement, the voltage across the sample would be varied while the current measured for each intervals of the voltage. The I/V is calculated from the slope of the I-V graph. Alternately, I/V could be calculated from the data using the least square method. Refer to the attachment 1 for more information on this least square method.

Energy gap $E_g$, of semiconducting and insulating materials could be determined from the variation of conductivity as a function of temperature. This is done by obtaining the I-V characteristics of the material (a series of I-V measurement) at varied temperature set-points. Thus, the conductivity at each temperature set-point could be calculated and $E_g$ calculated according to the equation:

$$\sigma = \sigma_o e^{-\frac{E_g}{2kT}}$$

Where $\sigma_o$ is the pre-exponential factor

K is Boltzmann's constant

T is the temperature in Kelvin

From the equation above:

$$\ln \sigma = \ln \sigma_o - \frac{E_g}{2k} \frac{1}{T}$$

Thus, $E_g$ could be calculated from the slope of ln $\sigma$ vs 1/T curve from the data obtained using the least square method.

**Problem:**

You are given a list of files each containing the I-V measurement data for a sample at temperatures varied from 0 – 100 °C with intervals of 20 °C. The files are named after the corresponding temperature set-point, i.e. file name **20** refers to the I-V data measured at 20 °C. Each file has 10 data points (10 measured I-V points). Please note that the temperature in the equation is in Kelvin. Below is the dimension of the sample:

| | | |
|---|---|---|
| Length | = | 2cm |
| Height (thickness) | = | 0.01 cm |
| Width | = | 1 cm |

**Assignment:**

Write a program to calculate the conductivity and energy gap of a material that would be able to process the files in the problem stated above. The program should be able to do the following:

(i) To open and read files pertaining to the data of I-V measurement.
(ii) To calculate the conductivity of the sample at each temperature intervals using the least square method for obtaining the slope of the I-V curve.
(iii) To calculate the energy gap of the material by using a ln $\sigma$ vs 1/T curve and using the least square method to determine the slope of the curve.
(iv) To save the calculation results in a separate file.

Advice: The use of function and array will significantly increase the efficiency and simplicity of the program.